

KARIGAR

Artisans, Amplified

Detailed Technical and Strategic Report AI-Driven Market Linkage and Smart Cataloging Application for Marginalized Artisans

SUPPLEMENTARY REPORT (COMPANION TO THE PRESENTATION)

Problem Statement ID	SIH26090
Theme	Culture & Heritage
Category	Software
Team	Team Karigar (Team ID 16)

Contents

Contents 1

1. Purpose and Scope of This Report 2

2. Sector Context and Evidence Base 2

3. Competitive Landscape and Differentiation 3

4. Users, Personas and Journeys 4

5. Detailed Functional Design 4

6. System Architecture 5

7. Data Model 6

8. AI Pipelines in Detail 7

9. Custom Order Workflow 8

10. Trust, Authenticity and Provenance 9

11. Security, Privacy and Compliance 9

12. Inclusive and Low-Resource Design 10

13. Business Model and Sustainability 10

14. Go-to-Market and Partnerships 11

15. Measurement and Evaluation Framework 11

16. Risks, Ethics and Mitigation 12

17. Detailed Implementation Roadmap 12

18. Assumptions, Limitations and Future Scope 13

19. Conclusion 13

Glossary 14

References 14

1. Purpose and Scope of This Report

The Team Karigar presentation for SIH26090 is deliberately compact: six slides cannot carry the design detail, evidence and planning that a serious solution needs. This report is the companion document. It records everything that could not be included in the slides, so that evaluators, mentors and future contributors can see how the idea would actually be built, launched, measured and sustained.

Where the presentation states *what* Karigar does, this report explains *how* it works, *why* the design choices were made, *who* it serves, *what it costs and earns*, and *what could go wrong*. Items that go beyond the presentation are the team's design proposals; they are labelled as proposed or illustrative wherever they are not drawn from a cited source.

Presentation slide	Expanded in this report
Problem (four barriers)	Section 2 (sector context), Section 4 (personas and journeys)
Solution (seven modules)	Section 5 (functional design), Section 9 (custom orders)
Technical approach	Sections 6, 7 and 8 (architecture, data model, AI pipelines)
Feasibility and viability	Sections 11, 12 and 13 (compliance, inclusive design, business model)
Impact and benefits	Section 15 (measurement framework)
Research and references	Section 3 (competitive landscape) and the reference list

2. Sector Context and Evidence Base

India has an estimated 64.66 lakh handloom and handicraft artisans, and women form 71% of handloom weavers and 64% of all artisans according to Ministry of Textiles data reported in August 2025 [1][10]. The Fourth All India Handloom Census 2019-20 found that 23.2% of rural handloom workers had never attended school [2]. These two facts define the design constraint for Karigar: a very large user base with limited formal schooling, for whom text-heavy interfaces are a barrier.

2.1 Existing identity and registration infrastructure

The PEHCHAN scheme, launched in 2016, issues Aadhaar-linked identity cards to handicraft artisans after verification by field functionaries of the Office of the Development Commissioner (Handicrafts). As of 30 June 2023, 31.14 lakh handicraft artisans had been registered [9]. This matters for Karigar because it offers a government-verified identity that can anchor artisan onboarding and authenticity claims, instead of the platform inventing its own verification from scratch.

2.2 Existing market-access initiatives

Initiative	What it offers	Gap Karigar addresses
ODOP (DPIIT)	1,243+ products across 775 districts, linking artisans to national and global markets [3]	Policy framework, not a tool that lets an individual artisan create listings
GeM ODOP Bazaar	Handicrafts across 446+ categories, connecting artisans to institutional buyers [4]	Procurement-oriented; onboarding and cataloguing still assume digital literacy
IndiaHandmade	Ministry marketplace; reported as zero-commission, with an Enrolment ID route for sellers without GST registration and a stated goal of onboarding 60 lakh+ artisans [10]	Listing creation, pricing help and custom orders are the artisan's own effort

Initiative	What it offers	Gap Karigar addresses
PEHCHAN card	Aadhaar-linked identity and access to scheme benefits [9]	Identity only; no commerce or cataloguing layer

The IndiaHandmade details are taken from a secondary summary of the Ministry's August 2025 announcement. The team should confirm them against the primary PIB release before final submission.

Key insight: the infrastructure for identity, policy support and marketplaces already exists. What is missing is the last mile, an assistant that lets a low-literacy artisan turn a photo and a spoken sentence into a complete, fairly priced, verifiable listing. This is where Karigar should position itself.

3. Competitive Landscape and Differentiation

An honest positioning analysis is necessary because the presentation proposes a commission-based model while a government marketplace is reported to charge sellers no commission [10]. Karigar therefore cannot compete on fees alone. Its case rests on capability, not price.

Category	Typical strengths	Typical limitations for low-literacy artisans	Karigar's angle
General e-commerce marketplaces	Large buyer base, logistics, payments	Listing tools assume typing, English and product-photography skills; built for ready-made goods	Voice-first listing; custom-order workflow
Government portals (GeM, IndiaHandmade)	Trust, zero or low fees, policy backing	Manual listing effort; limited AI assistance	AI cataloging, pricing help and buyer matching on top of the same ecosystem
Social media selling (WhatsApp, Instagram)	Free, familiar	No inventory, payment protection or provenance; heavy reliance on intermediaries	Structured catalogue, escrow, verified provenance
Artisan-focused brands and NGOs	Curation, storytelling	Limited capacity; onboard few artisans	Scalable onboarding through assisted mode

Table entries are the team's qualitative positioning analysis, not measured data. Specific competitor features should be verified before being quoted in a presentation.

3.1 Strategic options

Two strategies follow from this analysis. First, **complement rather than replace** government platforms: Karigar can act as the assisted creation and pricing layer and, in future, publish listings to GeM, IndiaHandmade or ONDC-compatible channels where technically and legally permitted. Second, **differentiate on custom orders and provenance**, which the presentation identifies as gaps in conventional e-commerce.

4. Users, Personas and Journeys

The following personas are illustrative composites created for design purposes. They are not real individuals and are not drawn from survey data.

Persona	Profile	Primary need	Design implication
Artisan (first-time digital user)	Rural weaver or potter, low schooling, basic smartphone, speaks a regional language	List a product without typing; get a fair price; receive payment safely	Voice-first flow, icons, audio confirmations, assisted mode
Artisan (semi-digital)	Uses WhatsApp, sells at fairs, some schooling	Reach buyers outside the district; manage stock	Bulk upload, inventory, analytics
Individual buyer (B2C)	Urban or overseas buyer valuing authentic craft	Verify maker and origin; request customisation	Provenance page, in-app chat, translation
B2B / institutional buyer	Boutique, exporter, emporium or department	Reliable supply, bulk pricing, custom specifications	Bulk quotes, verified artisan tiers, order milestones
Field agent / CSC operator	Helps artisans register and list	Onboard many artisans efficiently	Assisted mode with consent capture and audit trail

4.1 Example journey: first listing

- The artisan opens the app, chooses their language by tapping a flag-free language tile with a spoken label, and registers with a mobile number (or with help from a CSC operator).
- They photograph a finished product and press a large microphone button to describe it in their own words, for example the craft, the material, the time taken and the size.
- The app returns a spoken and visual summary of the draft listing. The artisan confirms or corrects it by voice.
- A suggested price band appears with a plain-language reason. The artisan sets the final price.
- The listing is published, and the artisan receives an audio notification when a buyer enquires.

5. Detailed Functional Design

The seven modules shown on the presentation slide are specified below in more depth. Requirements are grouped by module; each is written so it can be tested.

Module	Detailed functions	Key design rules
1. Onboarding	Mobile OTP sign-up; profile with craft, cluster, years of practice; optional PEHCHAN or Enrolment ID linkage; bank or UPI setup; language selection; consent capture	Never require typing to complete registration; allow agent-assisted mode with recorded consent
2. Smart cataloging	Photo capture guidance (lighting and framing hints); voice description; AI-generated title, category, description, tags, craft style, origin; suggested price range; SEO-friendly text; multi-language versions	AI output is always a draft; the artisan must confirm before publishing
3. Inventory and listing	Stock count; variants (size, colour, design); bulk upload; low-stock alerts; listing preview; draft and archive states	Simple counters and icons; offline drafts sync when connectivity returns

Module	Detailed functions	Key design rules
4. Dynamic pricing	Cost floor from materials and labour; market band from comparable listings; seasonal adjustments; explanation of each suggestion	Never suggest a price below the artisan's cost floor; artisan always has final say
5. Market linkage hub	Matching to B2C, B2B and institutional buyers; buyer discovery by craft, region and price; government marketplace and fair participation (future integration)	Matching explains why a buyer is shown; artisan can opt out of channels
6. Custom orders	Buyer request with reference images; chat with translation; quote and design confirmation; milestone payments; change requests	See Section 9
7. Analytics	Sales and revenue dashboard; top products; buyer regions; demand signals; promotion tools and social sharing	Charts use icons and few numbers; audio summary available

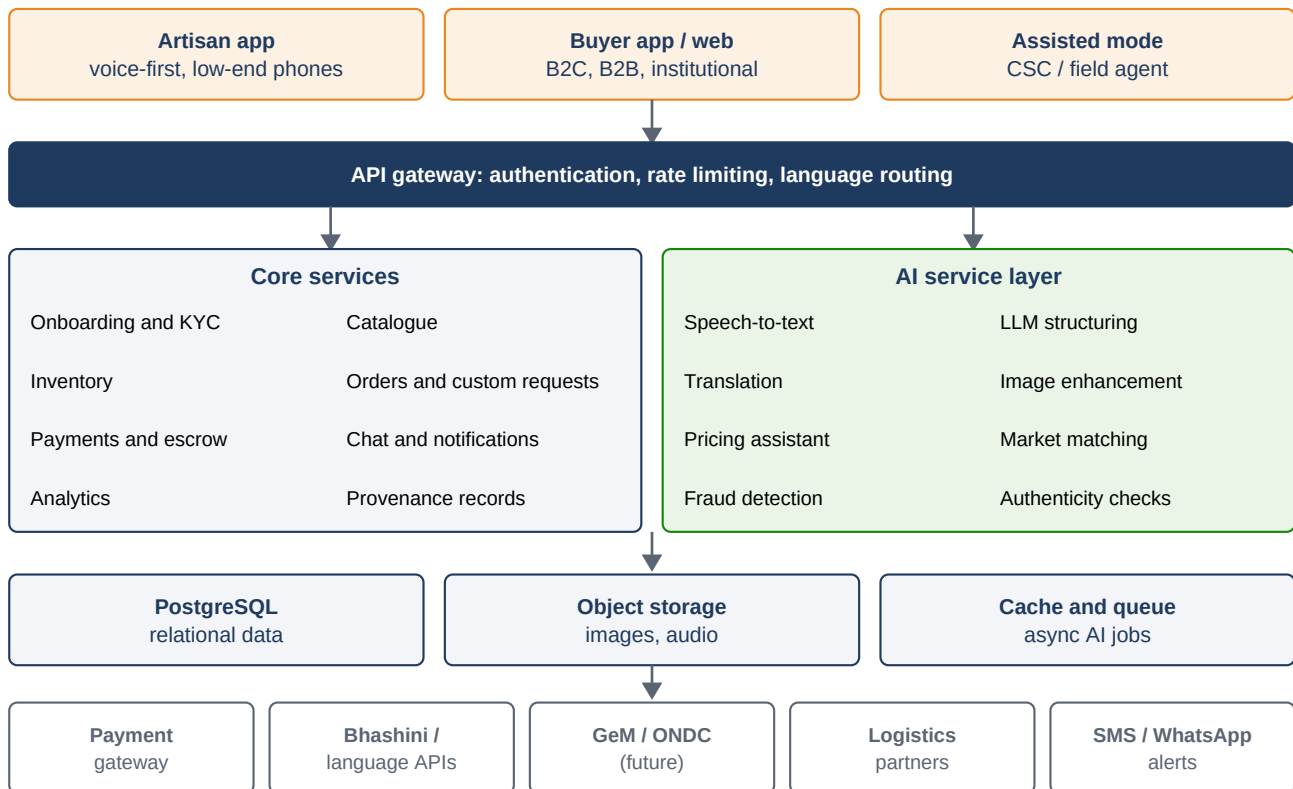
5.1 Buyer-side features

- Search and filter by craft, region, material, price and verification level.
- Provenance page for every product showing maker, cluster, craft story and verification tier.
- Translated descriptions and chat, so buyers and artisans can communicate without a shared language.
- Order tracking, purchase protection and a dispute channel.

6. System Architecture

The architecture separates client applications, core commerce services, an AI service layer and data storage, connected through a single API gateway. Keeping AI functions as separate services means they can be replaced or upgraded (for example, swapping one speech model for another) without touching commerce logic.

Figure 1. Layered system architecture



6.1 Design principles

- **Thin client, heavy cloud.** Heavy AI processing runs on the server so that a basic 2 GB RAM phone only needs to capture media and display results.
- **Asynchronous AI jobs.** Speech, image and language tasks are queued so the app stays responsive on slow networks; the artisan receives a notification when the draft is ready.
- **Graceful degradation.** If an AI service fails, the artisan can still list manually or with agent help.
- **Human in the loop.** No AI output is published without explicit artisan confirmation.

7. Data Model

The relational model below uses PostgreSQL, as stated in the technical approach. Only principal entities and key attributes are listed; audit columns and indexes are omitted for brevity.

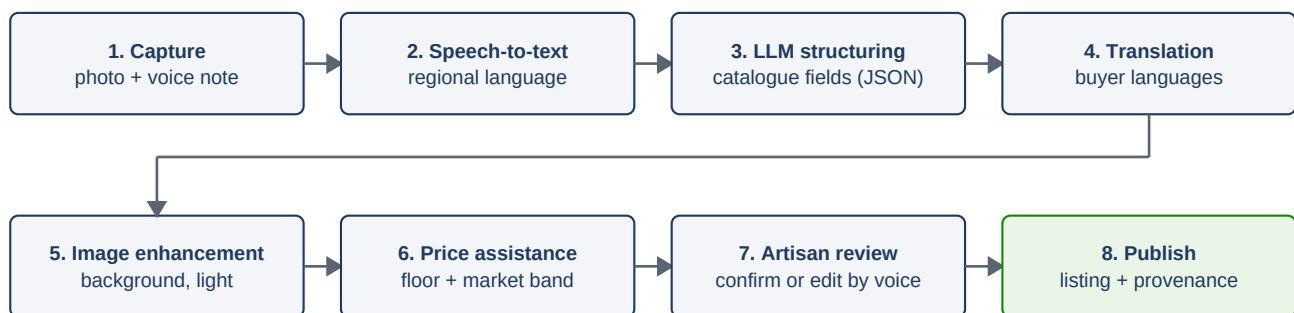
Entity	Key attributes	Notes
Artisan	id, name, phone, language, craft, cluster, verification_tier, payout_account_ref	Aadhaar number is not stored; only a verification status and reference
Product	id, artisan_id, title, description, category, craft_style, origin, materials, cost_floor, status	Descriptions stored per language in a translation table
Variant and Stock	product_id, size, colour, design_code, quantity, price	Supports bulk update
Media	id, product_id, type (photo, audio), original_uri, enhanced_uri, transcript	Originals kept so enhancement is reversible
Order	id, buyer_id, artisan_id, type (ready-made, custom), status, total, escrow_status	Status follows the state machine in Section 9

Entity	Key attributes	Notes
CustomRequest	order_id, requirements, reference_media, agreed_spec, milestones	Versioned so changes are traceable
Message	id, order_id, sender, original_text, translated_text, language	Both versions stored for dispute resolution
ProvenanceRecord	product_id, maker, cluster, technique, materials_source, craft_story, verification_tier, qr_code_ref	Public page rendered from this record
Payment	id, order_id, amount, method, status, gateway_ref	No card data stored on Karigar servers
Review and Report	id, order_id, rating, text, flags	Feeds fraud and quality signals

8. AI Pipelines in Detail

Figure 2 shows the core voice-and-photo flow. Each stage is described below, with its inputs, outputs and known risks.

Figure 2. Voice-and-photo to listing pipeline



8.1 Speech-to-text

The presentation names Whisper for converting artisan speech into text. Accuracy of general-purpose speech models varies considerably across Indian languages, dialects and noisy rural recording conditions, so the team proposes a **hybrid approach**: evaluate Whisper alongside India-specific speech services such as those available through the BHASHINI platform, choose per language on measured error rates, and keep the option of fine-tuning with crowd-sourced artisan speech. BHASHINI is the national language platform under MeitY; a July 2026 summary reports support for 36 Indian text languages and 23 voice languages [11].

8.2 LLM structuring

The transcript, together with photo-derived hints, is passed to a large language model with a strict output schema (title, category, materials, dimensions, craft style, origin, care instructions, description). The prompt instructs the model to use only facts the artisan stated or that are visible in the image and to leave a field empty rather than guess. Empty or low-confidence fields are highlighted so the artisan can fill them by voice.

8.3 Translation

Descriptions are translated into the buyer's language, and chat messages are translated in both directions. Craft terms (for example weave names or motif names) are kept in a glossary so they are transliterated

rather than mistranslated. Both original and translated text are stored.

8.4 Image enhancement

The pipeline removes cluttered backgrounds, corrects lighting and improves resolution. Enhancement must not alter the product's true colour, pattern or proportions; a side-by-side view is shown to the artisan, and the original is retained. A capture-time guide (for example, a prompt to move to daylight) reduces the amount of correction needed.

8.5 Pricing assistant

The assistant produces a suggested band in two steps. First it computes a **cost floor** from materials, labour hours multiplied by a fair hourly rate, and overhead. Then it estimates a **market band** from comparable listings, demand signals and seasonality. The recommendation is never below the cost floor, and every suggestion carries a plain-language reason, for example "similar hand-woven items sell in this range". The purpose is fair pricing for the artisan, not price undercutting.

8.6 Market-matching engine

Matching combines content-based signals (craft, material, price, region) with buyer behaviour (searches, saved items, past orders) and, for B2B, stated requirements such as volume and lead time. Buyers see why an item was recommended, and artisans can see which buyer segments are viewing their work. In the early phase, when data is scarce, matching relies on simple rules and popularity by category.

8.7 Fraud and anomaly detection

Rules and statistical anomaly detection flag suspicious patterns: duplicate or stolen images, sudden bulk listings from a new account, abnormal order or refund behaviour, and price outliers. Flagged cases go to a human reviewer; automated blocking is limited to high-confidence cases.

8.8 Authenticity assistance

AI can help detect duplicate or reused images and inconsistent claims, but **AI alone cannot certify that an item is handmade or from a given place**. Karigar therefore uses tiered verification, described in Section 10, in which AI signals support, but never replace, identity checks and human attestation.

9. Custom Order Workflow

Custom orders are one of the four barriers identified in the presentation. The workflow below lets a buyer and an artisan agree on a made-to-order item while protecting both sides.

Stage	What happens	Protection
1. Request	Buyer describes the need and attaches reference images; the system translates it for the artisan	Request is stored and versioned
2. Clarify	Artisan and buyer chat in their own languages; artisan may reply by voice note (transcribed and translated)	Both original and translated messages are kept
3. Quote	Artisan proposes price, size, colour, timeline; the pricing assistant checks the cost floor	Quote is an itemised, confirmable object
4. Confirm	Buyer accepts the agreed specification; advance payment is held in escrow	Funds are not released to either side yet
5. Make	Artisan shares progress photos at agreed milestones	Milestone approvals can release part of the payment

Stage	What happens	Protection
6. Deliver	Shipping with tracking; buyer confirms receipt	Escrow releases on confirmation or after a defined review window
7. Resolve	Any dispute is reviewed against the agreed specification and the message and photo history	Evidence trail supports fair outcomes

10. Trust, Authenticity and Provenance

Customers cannot easily verify who made a product or where it came from. Karigar addresses this with a provenance record attached to each product and a tiered verification model. The tiers are a team proposal.

Tier	Basis	What the buyer sees
Tier 1: Self-declared	Mobile-verified artisan with self-reported craft and origin	"Maker-declared" label
Tier 2: ID-verified	Artisan identity linked to a government artisan identity such as PEHCHAN or an Enrolment ID, subject to available integration	"ID-verified artisan" label
Tier 3: Attested	Craft and origin confirmed by a cluster body, cooperative, Development Commissioner (Handicrafts) office or another recognised institution; GI tag recorded where applicable	"Attested craft" label with attesting body named

Each provenance page can include the maker's name and photograph (with consent), the cluster and region, the technique, material sources, the time taken and a short craft story recorded in the artisan's own voice. A QR code on the product or packaging links to this page, which supports the cultural documentation goal in the presentation.

11. Security, Privacy and Compliance

The following controls are proposed. Legal and regulatory requirements change, and the team should obtain professional legal advice before launch; this report is not legal advice.

Area	Proposed control
Personal data	Follow the principles of India's Digital Personal Data Protection Act, 2023: informed consent in the user's language, purpose limitation, data minimisation, and the ability to withdraw consent and delete data
Identity	Avoid storing Aadhaar numbers; store only verification status and a reference; use authorised verification routes only
Payments	Use RBI-authorised payment gateways; no card or bank credentials stored on Karigar servers; escrow through a compliant arrangement
Images and voice	Explicit consent for publishing a maker's face or voice; original media stored encrypted; artisans can remove their media
Access control	Role-based access (artisan, buyer, agent, reviewer, admin); agents act only with recorded artisan consent
Data in transit and at rest	TLS for all traffic; encrypted storage; regular backups; audit logs for sensitive actions
AI safety	Human confirmation before publication; logging of AI outputs and edits; bias and accuracy monitoring

Area	Proposed control
Marketplace rules	Terms for sellers and buyers, grievance redressal, returns policy, and compliance with applicable e-commerce and consumer protection rules

12. Inclusive and Low-Resource Design

The presentation states that even a basic 2 GB RAM phone can use the application. The following measures make that claim testable.

- **Voice-first interaction.** Every step can be completed by speaking and tapping; text entry is optional.
- **Icon-led interface.** Large touch targets, consistent icons, minimal text, and audio playback of any on-screen message.
- **Lightweight client.** Compressed images before upload, a small install size, and lazy loading of screens. A progressive web app or a lean native build should be benchmarked on a 2 GB device.
- **Low-bandwidth mode.** Upload queues that resume after a drop in connection, and offline draft saving.
- **Assisted mode.** A CSC operator or field agent can onboard and list on behalf of an artisan, with consent recorded.
- **Women artisans.** Because women form most of the handloom workforce [10], the design should account for shared phones, limited screen time and the option for a family member or self-help group to assist.
- **Usability testing.** Test with real artisans in each pilot language before scaling, and measure task completion without help.

13. Business Model and Sustainability

The presentation proposes a small commission on successful sales. Because a government marketplace is reported to charge no commission [10], the team should treat the commission as one option among several and test willingness to pay.

Revenue stream	Description	Comment
Small commission	A low percentage on completed sales, charged only when the artisan earns	Simple to explain; must remain visibly fair
B2B services	Lead-matching or sourcing fees from institutional and business buyers	Shifts cost away from artisans
Institutional partnerships	Licensing or sponsorship from state bodies, CSR programmes or NGOs that fund artisan onboarding	Suits assisted-mode rollouts
Value-added services	Optional premium analytics, promotion tools, packaging or logistics margins	Optional for artisans, never mandatory

13.1 Illustrative commission revenue

The table shows platform revenue from 1,000 completed orders under different assumptions of average order value and commission rate. The values are simple arithmetic on hypothetical inputs, not forecasts. Revenue equals orders multiplied by average order value multiplied by commission rate.

Average order value	3% commission	5% commission	8% commission
Rs. 1,000	Rs. 30,000	Rs. 50,000	Rs. 80,000
Rs. 2,000	Rs. 60,000	Rs. 1,00,000	Rs. 1,60,000
Rs. 3,000	Rs. 90,000	Rs. 1,50,000	Rs. 2,40,000

The main cost drivers to model are AI inference (speech, language and image processing), cloud storage and bandwidth, payment gateway fees, customer support and field operations for assisted onboarding. A pilot should measure the actual cost per listing created and cost per order fulfilled, then set fees so that the platform is sustainable while artisans keep a clearly higher share of the price than they would through intermediaries.

14. Go-to-Market and Partnerships

14.1 Pilot approach

The presentation cites Ghe Bharari in Maharashtra as an example of a fair-based sales channel. A Maharashtra pilot is therefore a natural starting point, working with a small number of craft clusters. Candidate crafts and clusters should be chosen with local partners; the team proposes selecting one textile craft and one non-textile craft so that both photography and description challenges are tested.

14.2 Partner ecosystem

Partner type	Role
Office of the Development Commissioner (Handicrafts), cluster bodies and cooperatives	Artisan identification, attestation, and field access
Common Service Centres and self-help groups	Assisted onboarding and digital literacy support
Design institutes (for example IAD)	Design guidance and cultural documentation
Payment and logistics providers	Escrow, payouts and delivery
Language platforms such as BHASHINI	Speech and translation services for Indian languages
Government marketplaces (GeM, IndiaHandmade)	Future channel integration for institutional buyers

15. Measurement and Evaluation Framework

Proposed targets are starting hypotheses to be validated in the pilot, not commitments.

Area	Metric	Definition	Proposed starting target
Usability	Time to first listing	Minutes from photo capture to published listing	Under 5 minutes for a first-time user
Usability	Unaided completion rate	Share of new artisans who finish a listing without help	Majority of pilot users
AI quality	Transcription error rate	Word error rate by language on artisan speech	Set per language after baseline testing
AI quality	Artisan edit rate	Share of AI-generated fields changed before publishing	Falling over time

Area	Metric	Definition	Proposed starting target
Market	Out-of-district sales share	Portion of orders from outside the artisan's district	Increasing versus baseline
Income	Price realised	Average price per item versus the artisan's prior channel	Higher than baseline
Trust	Dispute rate	Orders with a dispute per 100 orders	Low and falling
Heritage	Provenance coverage	Share of listings with Tier 2 or Tier 3 verification	Increasing quarter on quarter

16. Risks, Ethics and Mitigation

Risk	Why it matters	Mitigation
AI mistranslation or wrong description	Misrepresents the craft; harms trust	Artisan confirmation, craft-term glossary, feedback loop
Speech model errors for dialects	Frustrates first-time users	Per-language evaluation, fallback to typed or assisted entry, fine-tuning
Cultural misrepresentation by the LLM	Inaccurate origin or meaning claims	Constrain outputs to stated facts; expert review of craft stories
Design copying	Machine-made copies of traditional designs undercut artisans	Provenance, GI recording, image duplicate detection, takedown process
Price pressure	Recommendations could push prices down	Cost-floor rule and explanations; monitoring for downward drift
Digital exclusion	Users without smartphones or connectivity are left behind	Assisted mode, cluster-level access points
Data misuse	Sensitive personal and financial data	Consent, minimisation, encryption, audit logs
Platform dependence	Artisans rely on one channel	Portable data export; multi-channel integration
Sustainability	Low fees may not cover costs	Multiple revenue streams; institutional partnerships

17. Detailed Implementation Roadmap

Timelines are indicative and depend on team size and partner readiness.

Phase	Indicative duration	Deliverables	Exit criteria
0. Discovery	4 to 6 weeks	Artisan interviews; language and device survey; partner agreements	Validated personas and pilot cluster list
1. Prototype	8 to 10 weeks	Voice-to-listing flow in two or three languages; image enhancement; basic listing	Artisans complete a listing unaided in usability tests
2. Pilot	3 to 4 months	Pricing assistant; orders and escrow; assisted mode; provenance pages	Real orders fulfilled; measured time-to-list and edit rate
3. Marketplace	4 to 6 months	Market-matching; custom orders; analytics; fraud detection	Sustained buyer activity across B2C and B2B

Phase	Indicative duration	Deliverables	Exit criteria
4. Scale	Ongoing	More languages and states; marketplace integrations; partner programmes	Onboarding cost per artisan falling

18. Assumptions, Limitations and Future Scope

18.1 Assumptions and limitations

- Speech and language accuracy for some Indian languages and dialects may be lower than for major languages and must be measured before claims are made.
- Access to government identity and marketplace integrations depends on approvals and available interfaces.
- Revenue figures in this report are illustrative and require pilot validation.
- Regulatory requirements, especially for payments and data protection, must be confirmed with qualified advisors.
- Some ecosystem facts come from secondary summaries of official announcements and should be checked against primary sources.

18.2 Future scope

- Augmented-reality preview so buyers can see a product in their space.
- Integration with logistics, packaging and export documentation.
- Linkage to credit and scheme benefits for verified artisans.
- Voice-based assistant for order updates and business advice in regional languages.
- Craft-heritage archive built from provenance records and artisan-recorded stories.

19. Conclusion

The presentation sets out the idea; this report shows the depth behind it. Karigar is built on existing national infrastructure for identity, language and marketplaces, and focuses on the last-mile problem that those systems do not solve: helping a low-literacy artisan create, price and sell a product with a photograph and a few spoken sentences. Its differentiators are voice-first AI cataloging, fair-price assistance, custom orders with protected payments, and tiered provenance. The principal challenges, namely speech accuracy in regional languages, sustainable economics against zero-commission alternatives, and credible authenticity, are addressed through a hybrid AI approach, multiple revenue streams and tiered verification. A staged pilot will show which assumptions hold.

Glossary

Term	Meaning
ASR	Automatic speech recognition: converts spoken language to text
BHASHINI	National language platform under MeitY providing speech and translation services for Indian languages
CSC	Common Service Centre: local digital service outlet
Escrow	Payment held by a neutral party until agreed conditions are met
GI tag	Geographical Indication: legal recognition that a product originates from a specific region
GeM	Government e-Marketplace
LLM	Large language model
ODOP	One District One Product
PEHCHAN	Aadhaar-linked identity card scheme for handicraft artisans
Provenance	Documented history of who made a product, where and how

References

- [1] Press Information Bureau, Ministry of Textiles (2025): estimate of 64.66 lakh handloom and handicraft artisans in India.
- [2] Ministry of Textiles, Fourth All India Handloom Census 2019-20.
- [3] Department for Promotion of Industry and Internal Trade (DPIIT), Government of India: One District One Product (ODOP) scheme.
- [4] Government e-Marketplace (GeM): ODOP Bazaar.
- [5] Ministry of Textiles: IndiaHandmade.
- [6] Development Commissioner (Handicrafts), Government of India.
- [7] BharatGo Blog: How Local Artisans in India Can Expand Their Reach Through E-commerce.
- [8] Indian Institute of Art and Design (IIAD): The Village, A Journey Through Goan Culture and Culinary Traditions.
- [9] Lok Sabha Unstarred Question No. 2264, answered 2 August 2023: PEHCHAN Card Scheme (31.14 lakh handicraft artisans registered as on 30.06.2023).
- [10] Current-affairs summary of the Ministry of Textiles announcement on the IndiaHandmade platform, August 2025 (Ananta IAS): zero commission, Enrolment ID route, 71% of handloom weavers and 64% of artisans are women, target of 60 lakh+ artisans. Secondary source; verify against the primary release.
- [11] Drishti IAS, BHASHINI launches Rajasthan Language Model Training Hackathon, 4 July 2026: platform supports 36 Indian text languages and 23 Indian voice languages. Secondary source; verify at bhashini.gov.in.
- [12] Government of India: Digital Personal Data Protection Act, 2023 (general reference).